ID: 22426160

COURSE CODE: ASTA607

COURSE NAME: COMPUTATIONAL STATISTICS 1

END OF FIRST SEMESTER EXAMS

DEPT. OF STAT. & ACTU. SCIENCE

QUESTION 1

CODES:

###QUESTION 1

```
#(a)
packageDescription("MASS")

#(b)
data("crabs")
str(crabs)
summary(crabs)
head(crabs)

#(c)
freq_sp <- table(crabs$sp)
freq_sex <- table(crabs$sex)
freq_sp_sex <- table(crabs$sp, crabs$sex)

#(d)
barplot(freq_sp_sex,
        beside = TRUE,
        col = c(4:5),
```

```
        legend = rownames(freq_sp_sex),
        main = "Crab Group Composition",
        xlab = "Sex",
        ylab = "Count")
```

(e) The crabs dataset is perfectly balanced each species having 100 crabs and with 50 males and 50 females. This makes the data ideal for making for unbiased comparative analysis and decision making.

OUTPUT

###QUESTION 1

```
>

> #(a)

>

> packageDescription("MASS")
```

> #(b)

```
> data("crabs")

> str(crabs)

'data.frame': 200 obs. of  8 variables:

 $ sp   : Factor w/ 2 levels "B","O": 1 1 1 1 1 1 1 1 1 1
...

 $ sex  : Factor w/ 2 levels "F","M": 2 2 2 2 2 2 2 2 2 2
...

 $ index: int  1 2 3 4 5 6 7 8 9 10 ...

 $ FL   : num  8.1 8.8 9.2 9.6 9.8 10.8 11.1 11.6 11.8 11.8
...

 $ RW   : num  6.7 7.7 7.8 7.9 8 9 9.9 9.1 9.6 10.5 ...
```

```
 $ CL   : num  16.1 18.1 19 20.1 20.3 23 23.8 24.5 24.2
25.2 ...
 $ CW   : num  19 20.8 22.4 23.1 23 26.5 27.1 28.4 27.8
29.3 ...
 $ BD   : num  7 7.4 7.7 8.2 8.2 9.8 9.8 10.4 9.7 10.3 ...
> summary(crabs)
 sp      sex         index             FL               RW
CL
 B:100   F:100   Min.   : 1.0   Min.   : 7.20   Min.   :
6.50   Min.   :14.70
 O:100   M:100   1st Qu.:13.0   1st Qu.:12.90   1st
Qu.:11.00   1st Qu.:27.27
                 Median :25.5   Median :15.55   Median
:12.80   Median :32.10
                 Mean   :25.5   Mean   :15.58   Mean
:12.74   Mean   :32.11
                 3rd Qu.:38.0   3rd Qu.:18.05   3rd
Qu.:14.30   3rd Qu.:37.23
                 Max.   :50.0   Max.   :23.10   Max.
:20.20   Max.   :47.60
       CW              BD
 Min.   :17.10   Min.   : 6.10
 1st Qu.:31.50   1st Qu.:11.40
 Median :36.80   Median :13.90
 Mean   :36.41   Mean   :14.03
 3rd Qu.:42.00   3rd Qu.:16.60
 Max.   :54.60   Max.   :21.60
> head(crabs)
  sp sex index   FL  RW   CL   CW  BD
1  B   M     1  8.1 6.7 16.1 19.0 7.0
```

```
2  B   M      2  8.8 7.7 18.1 20.8 7.4
3  B   M      3  9.2 7.8 19.0 22.4 7.7
4  B   M      4  9.6 7.9 20.1 23.1 8.2
5  B   M      5  9.8 8.0 20.3 23.0 8.2
6  B   M      6 10.8 9.0 23.0 26.5 9.8
>
```

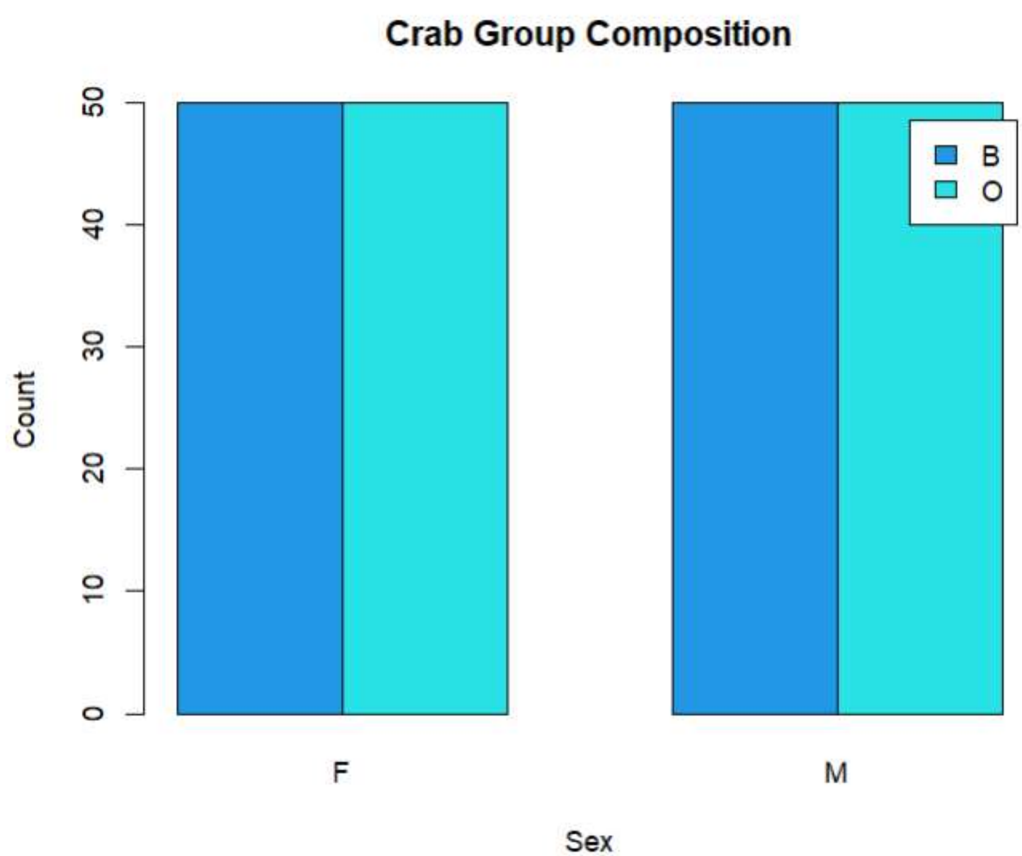
> #(c)

```
>
> freq_sp <- table(crabs$sp)
> freq_sex <- table(crabs$sex)
> freq_sp_sex <- table(crabs$sp, crabs$sex)
```

> #(d)

```
> barplot(freq_sp_sex,
+         beside = TRUE,
+         col = c(4:5),
+         legend = rownames(freq_sp_sex),
+         main = "Crab Group Composition (MASS::crabs)",
+         xlab = "Sex",
+         ylab = "Count")
```

Figure 1.1



QUESTION 2

CODES

#(a)

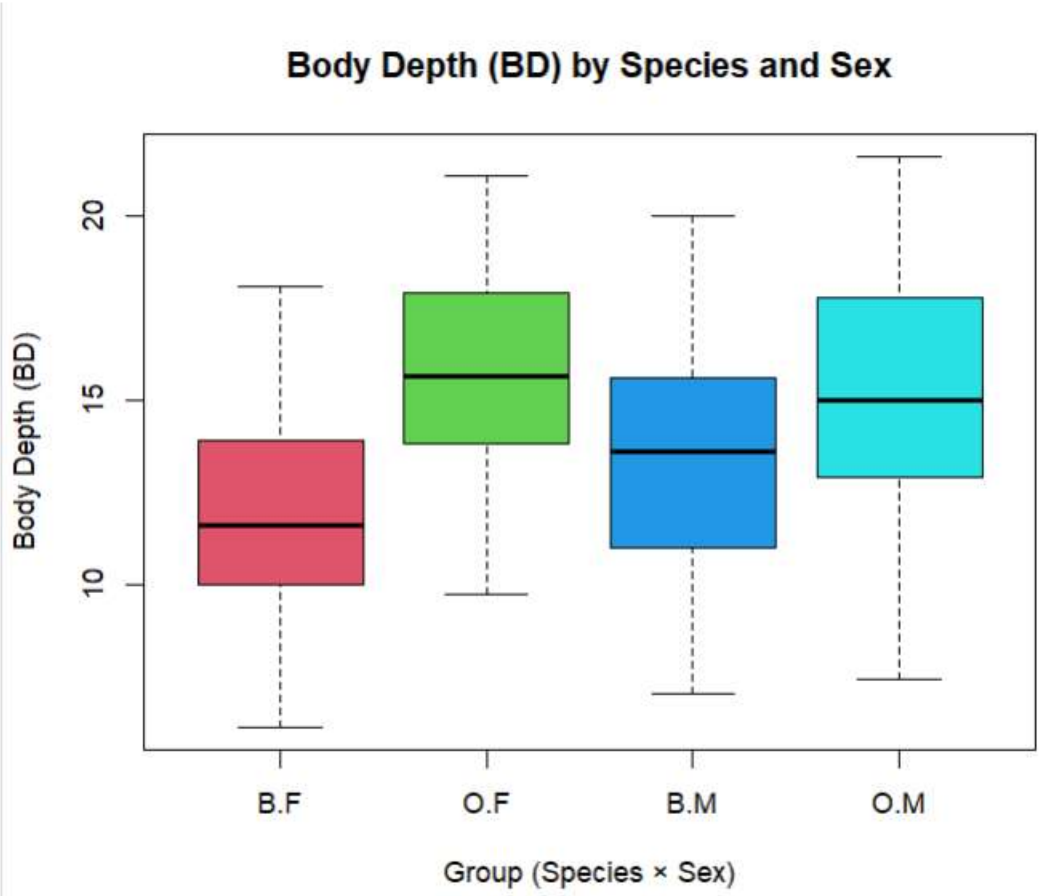
```
boxplot(BD ~ group, data = crabs,
        col = c(2:6),
        main = "Body Depth (BD) by Species and Sex",
        xlab = "Group (Species × Sex)",
        ylab = "Body Depth (BD)")
```

 # (b)

```
(cent_med<-tapply(crabs$BD, crabs$group, median))
(IQR_dat<-tapply(crabs$BD, crabs$group, IQR))
(SD_dat<-tapply(crabs$BD, crabs$group, sd))
```

OUTPUT

#(a) Figure 1.2



 # (b)

```
> (box_plot$out)
numeric(0)
>
> (cent_med<-tapply(crabs$BD, crabs$group, median))
  B.F   O.F   B.M   O.M
```

```
11.60 15.65 13.60 15.00
>
> (IQR_dat<-tapply(crabs$BD, crabs$group, IQR))
  B.F   O.F   B.M   O.M
3.850 4.050 4.600 4.825
>
> (SD_dat<-tapply(crabs$BD, crabs$group, sd))
     B.F      O.F      B.M      O.M
2.752465 2.752620 3.199888 3.527187
```

Comment: From figure 1.2 and the computed summaries, the boxplot of body depth shows that male crabs tend to be larger than females, and crabs of species O are slightly bigger than those of species B. There are a few unusually large individuals, mostly among the males. Overall, males show a bit more variation in body depth, while females are more consistent. The plot clearly highlights the differences in size and spread between the four groups. Outliers not visible.

QUESTION 3

CODES

```
> #(a)
> par(mfrow = c(1,1))
> bcl_1 <- crabs$CL[crabs$sp == "B"]
> ocl_1 <- crabs$CL[crabs$sp == "O"]
>
> par(mfrow = c(1, 2))
>
> hist(bcl_1,
+      col = 3,
+      main = "Carapace Length (CL) - Species B",
+      xlab = "CL",
+      ylab = "Frequency",
+      breaks = 10)
>
> hist(ocl_1,
+      col = 4,
+      main = "Carapace Length (CL) - Species O",
+      xlab = "CL",
+      ylab = "Frequency",
+      breaks = 10)
```

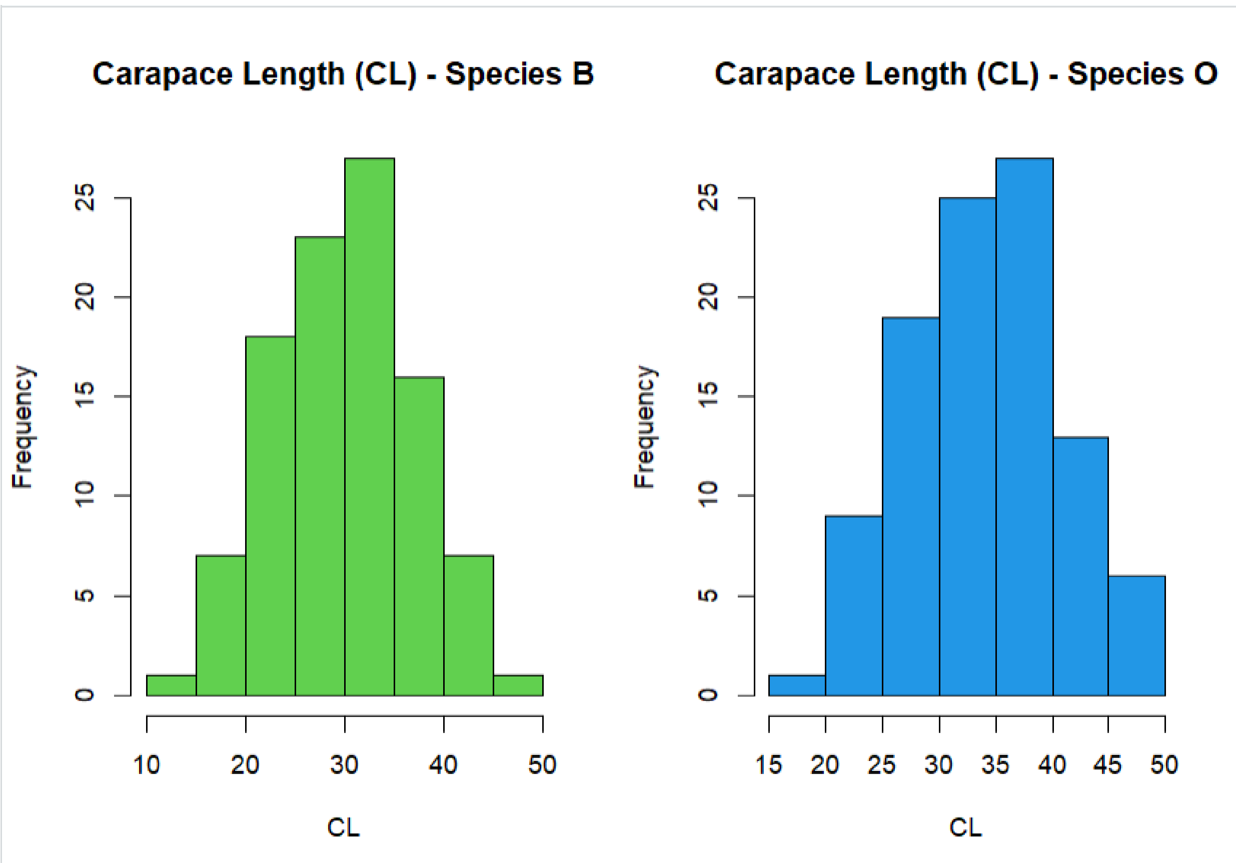
>
> #(b)

The histograms (figure 1.3) show that both species have a single peak in carapace length, so they are unimodal. Species B has a slight tendency toward smaller sizes, with a few cra bs on the lower end, while species O is more evenly spread and fairly symmetric. Overall, species O crabs are a little larger on average, but both species show a similar general patte rn in their distributions.

OUTPUT

#(a)

Figure 1.3



QUESTION 4

CODES

#(a)

```
mor_var <- crabs[, c("FL", "RW", "CL", "CW", "BD")]

(sp_mean <- aggregate(mor_var, by = list(Species = crabs$sp),
FUN = mean))

(sex_mean <- aggregate(mor_var, by = list(Sex = crabs$sex),
FUN = mean))

(gp_mean <- aggregate(mor_var,

          by = list(Species = crabs$sp,

          Sex = crabs$sex), FUN = mean))
```

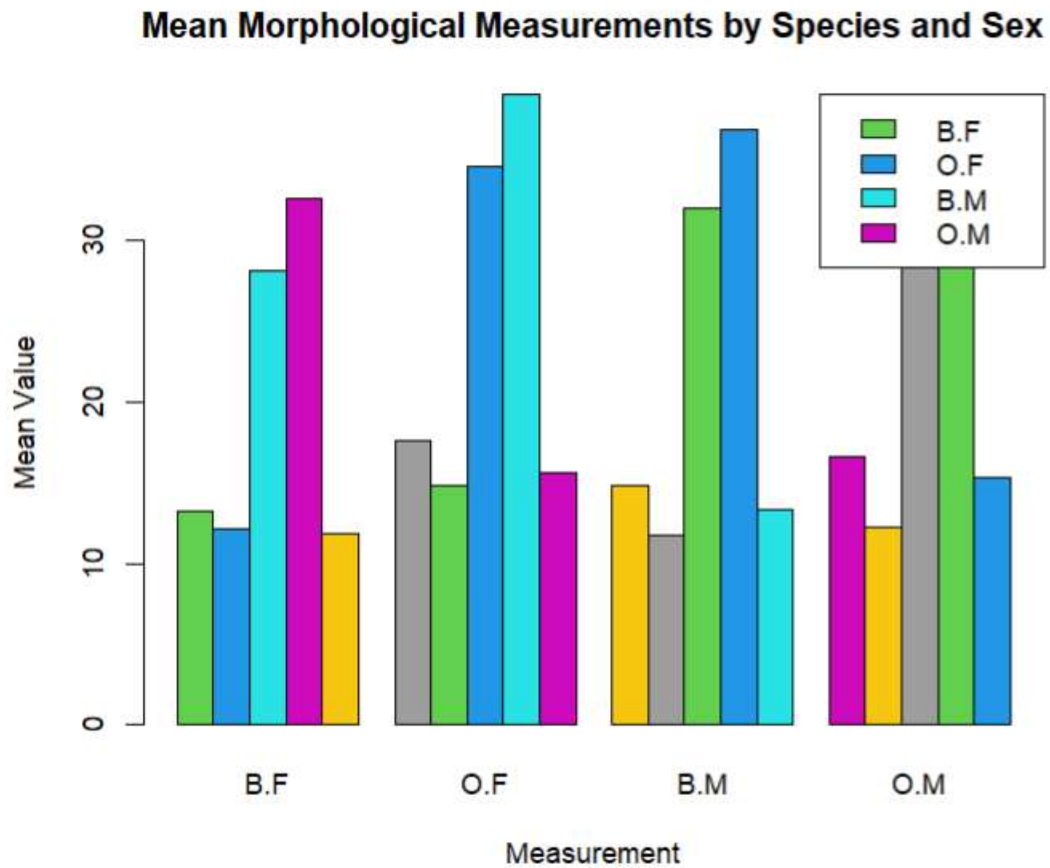
#(b)

```
mean_matx <- t(as.matrix(gp_mean[, -c(1,2)]))

colnames(mean_matx) <- paste(gp_mean$Species, gp_mean$Sex,
sep = ".")


barplot(mean_matx,

        beside = TRUE,

        col = c(3:8),

        main = "Mean Morphological Measurements by Species
and Sex",

        ylab = "Mean Value",

        xlab = "Measurement",

        legend = colnames(mean_matx),

        args.legend = list(x = "topright"))
```

OUTPUT

```
> #(a)
>
> mor_var <- crabs[, c("FL", "RW", "CL", "CW", "BD")]
>
> (sp_mean <- aggregate(mor_var, by = list(Species = crabs$
sp), FUN = mean))
  Species     FL     RW     CL     CW     BD
1       B 14.056 11.928 30.058 34.717 12.583
2       O 17.110 13.549 34.153 38.112 15.478
>
> (sex_mean <- aggregate(mor_var, by = list(Sex = crabs$sex
), FUN = mean))
  Sex     FL     RW     CL     CW     BD
1   F 15.432 13.487 31.360 35.830 13.724
2   M 15.734 11.990 32.851 36.999 14.337
>
> (gp_mean <- aggregate(mor_var,
+             by = list(Species = crabs$sp,
+             Sex = crabs$sex), FUN = mean))
  Species Sex     FL     RW     CL     CW     BD
1       B   F 13.270 12.138 28.102 32.624 11.816
2       O   F 17.594 14.836 34.618 39.036 15.632
3       B   M 14.842 11.718 32.014 36.810 13.350
4       O   M 16.626 12.262 33.688 37.188 15.324


#(b)
>
> mean_matx <- t(as.matrix(gp_mean[, -c(1,2)]))
> colnames(mean_matx) <- paste(gp_mean$Species, gp_mean$Sex
, sep = ".")
>
> barplot(mean_matx,
+          beside = TRUE,
+          col = c(3:8),
+          main = "Mean Morphological Measurements by Specie
s and Sex",
+          ylab = "Mean Value",
+          xlab = "Measurement",
+          legend = colnames(mean_matx),
+          args.legend = list(x = "topright"))
```

Figure 1.4



#(c) Comment:

Carapace width stands out as the measurement that differs most between species, with sp ecies O crabs being wider than species B. At the same time, it shows the smallest differen ce between males and females, meaning width is fairly similar within each species. Other measurements like carapace length and frontal lobe size show some differences, but widt h is the clearest marker for species while being the most consistent between sexes.

QUESTION 5

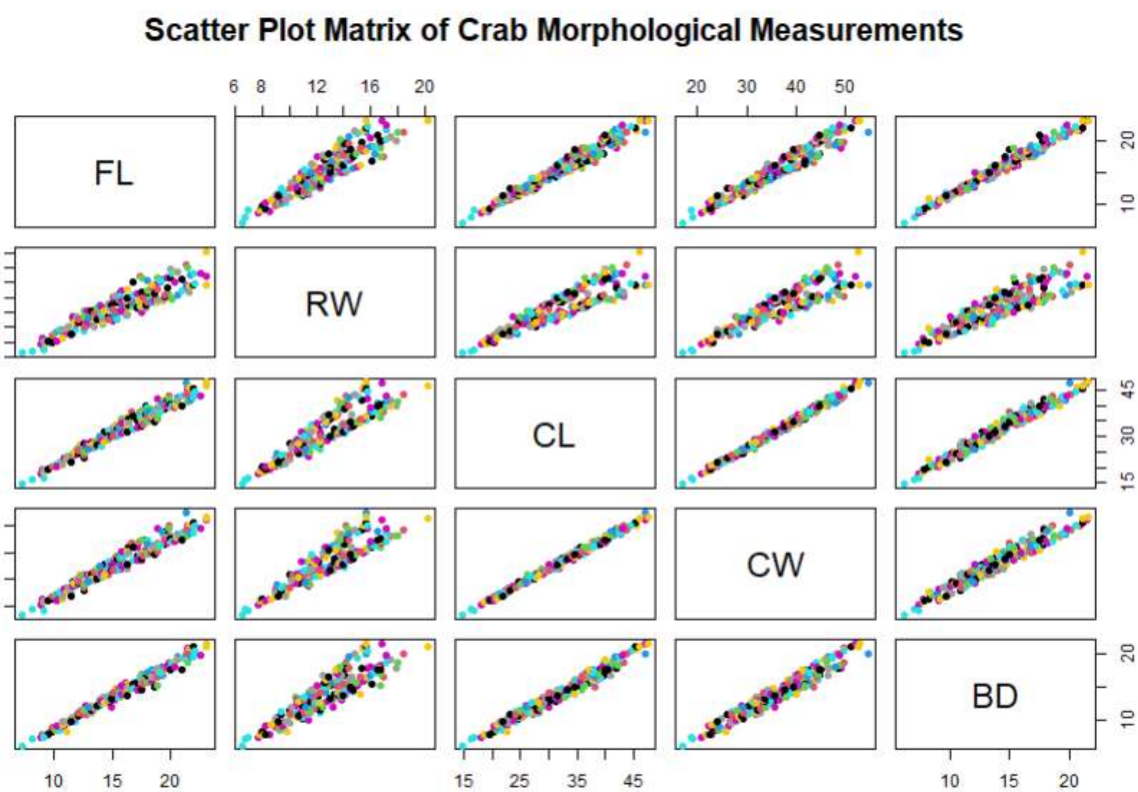
CODES

#(a)

```
pairs(mor_var,
      main = "Scatter Plot Matrix of Crab Morphological Measurements",
      pch = 19,
      col = 5:25)
```

OUTPUT

#(a) Figure 1.5



#(b) Comment:

In figure 1.5, the scatter plot matrix shows that all five morphological measurements increase together, meaning crabs that are larger in one measurement are generally larger in the others. The strongest relationships are seen between carapace length and carapace width, and between frontal lobe size and rear width, where the points cluster closely along an upward trend. Most of these relationships appear roughly linear, suggesting that increases in one dimension are proportional to increases in others. There is also a clear distinction between species: crabs of species O generally have larger measurements across all body parts compared with species B. While there is some overlap, the overall pattern shows that size differences between crabs are consistent across the five measurements. This indicates that the body parts grow together, and the measurements are closely linked, reflecting the overall size of each crab.

QUESTION 6

Report of Statistical Analysis of Morphological Measurements in Crabs

Introduction

This analysis utilizes the crabs dataset from the MASS package which contains measurements of 200 crabs, classified by species (B and O) and sex (F and M). Among the recorded variables, five key morphological measurements: Frontal Lobe size (FL), Rear Width (RW), Carapace Length (CL), Carapace Width (CW), and Body Depth (BD): offer insight into the overall size and body structure of the crabs. This report explores patterns in these measurements to understand differences between species and sexes, assess the relationships among the measurements, and identify which traits are most distinctive.

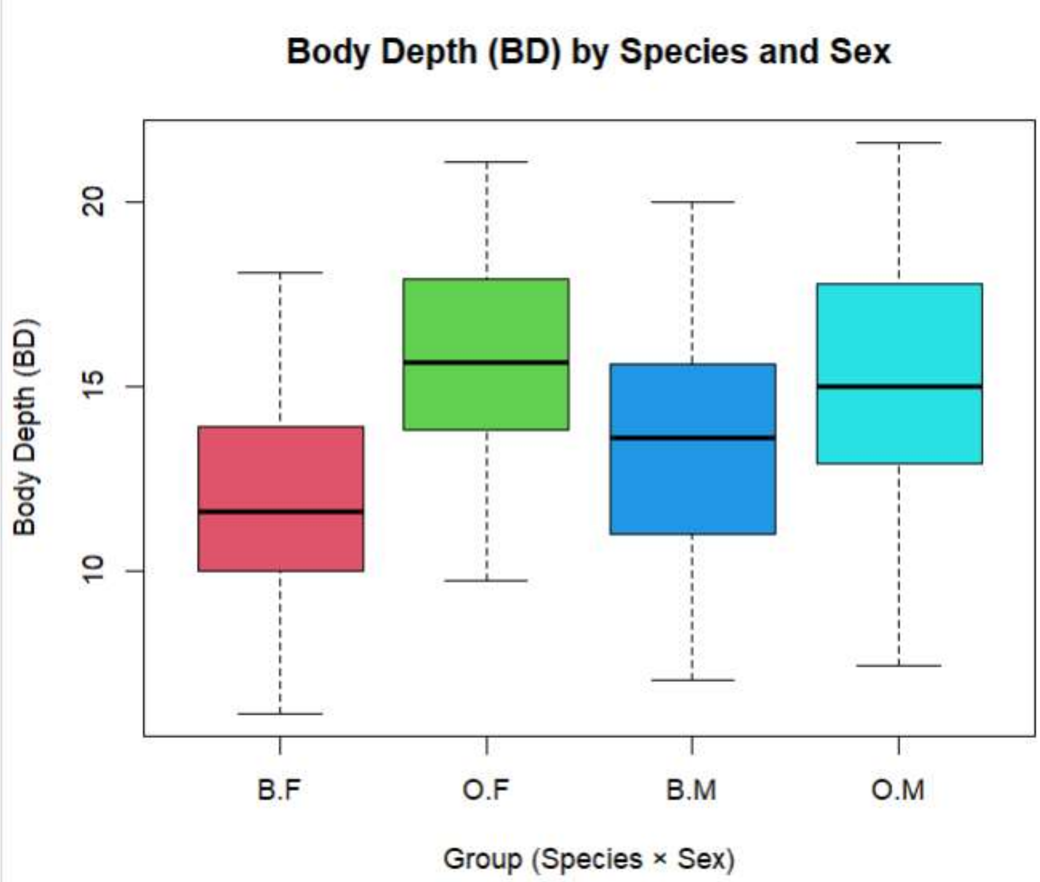
Methodology

The dataset was analyzed using R Studio where descriptive statistics was carried out to determine the outcome. In fact, a look at the dataset shows that the distribution of crabs is perfectly balanced. There are 100 individuals of each species, and within each species, there are 50 males and 50 females. A bar chart of species and sex confirms this, with equal counts across all four groups: B.F, B.M, O.F, and O.M. This balance is important because it ensures that comparisons of size and shape are not influenced by unequal sample sizes, allowing any differences we observe to reflect true biological variation.

Findings

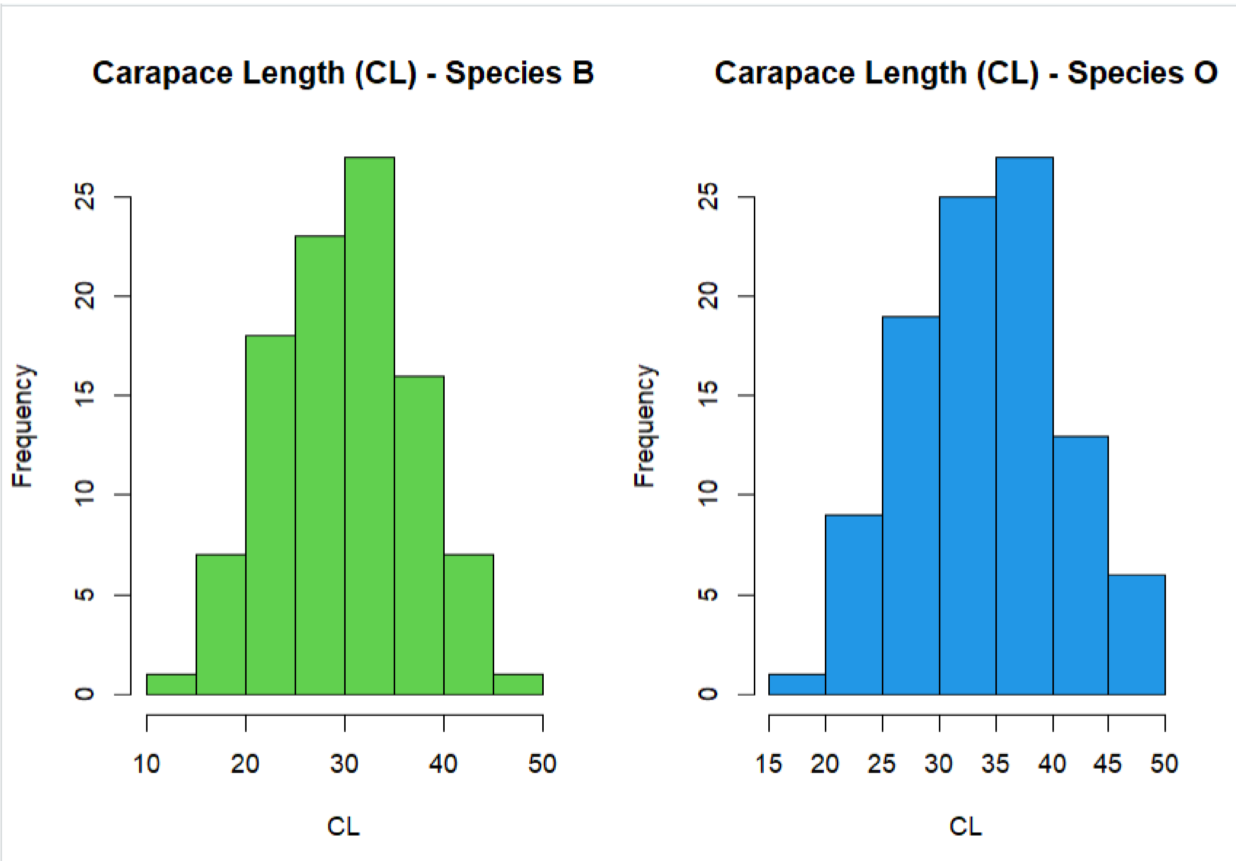
Based on the analysis carried out in R Studio, the following findings were observed. Firstly, the Body depth (BD) was examined across species and sex using a side-by-side boxplot. From figure 1.1, the results indicate that male crabs tend to be deeper-bodied than females, and crabs of species O are generally larger than those of species B. A few extreme values appear among the males, representing particularly large individuals. In addition, male crabs show slightly more variability in body depth, while females are more consistent. Overall, these patterns suggest that both species and sex affect body depth, with male O crabs being the largest and female B crabs being the smallest.

Figure 1.1



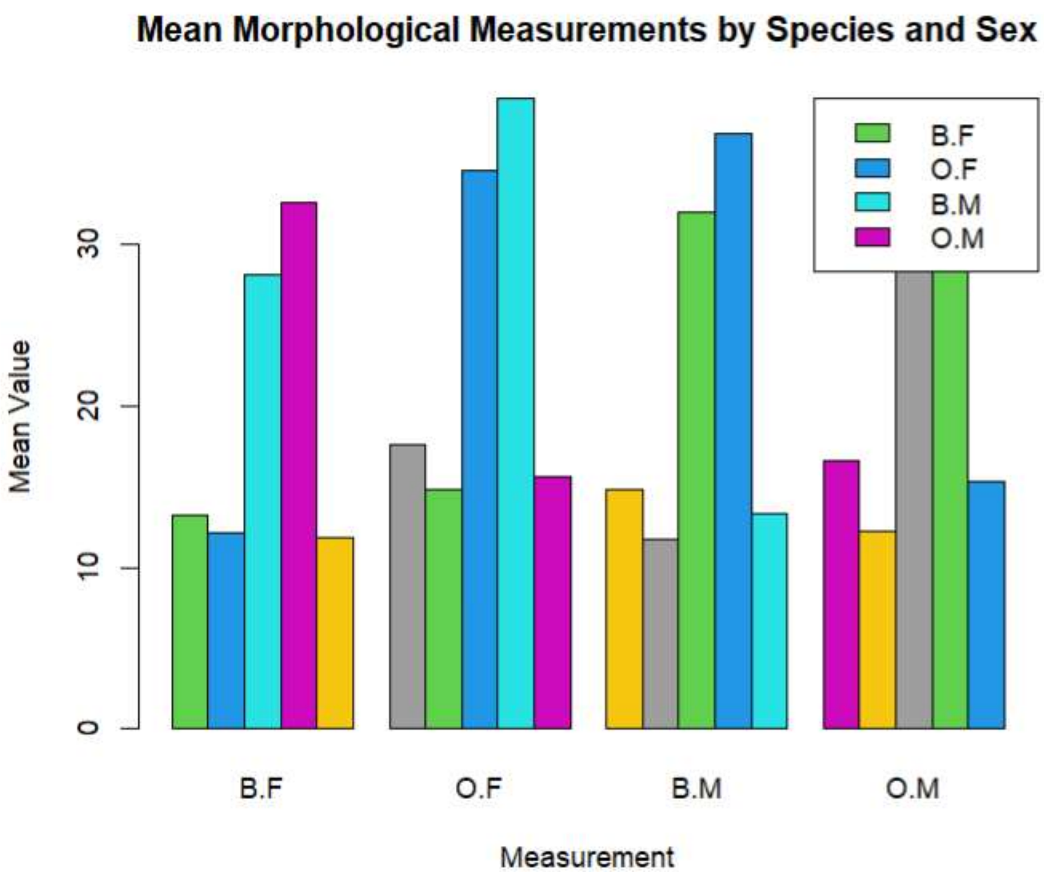
Also, a histograms (figure 1.2) of carapace length reveal that both species have unimodal distributions, each with a clear peak around the average size. Species B shows a slight skew toward smaller individuals, while species O is more symmetric and centered at a higher value. On average, species O crabs have longer carapaces than species B. These plots demonstrate that, while both species share a similar general distribution, species O crabs are consistently larger in length.

Figure 1.2



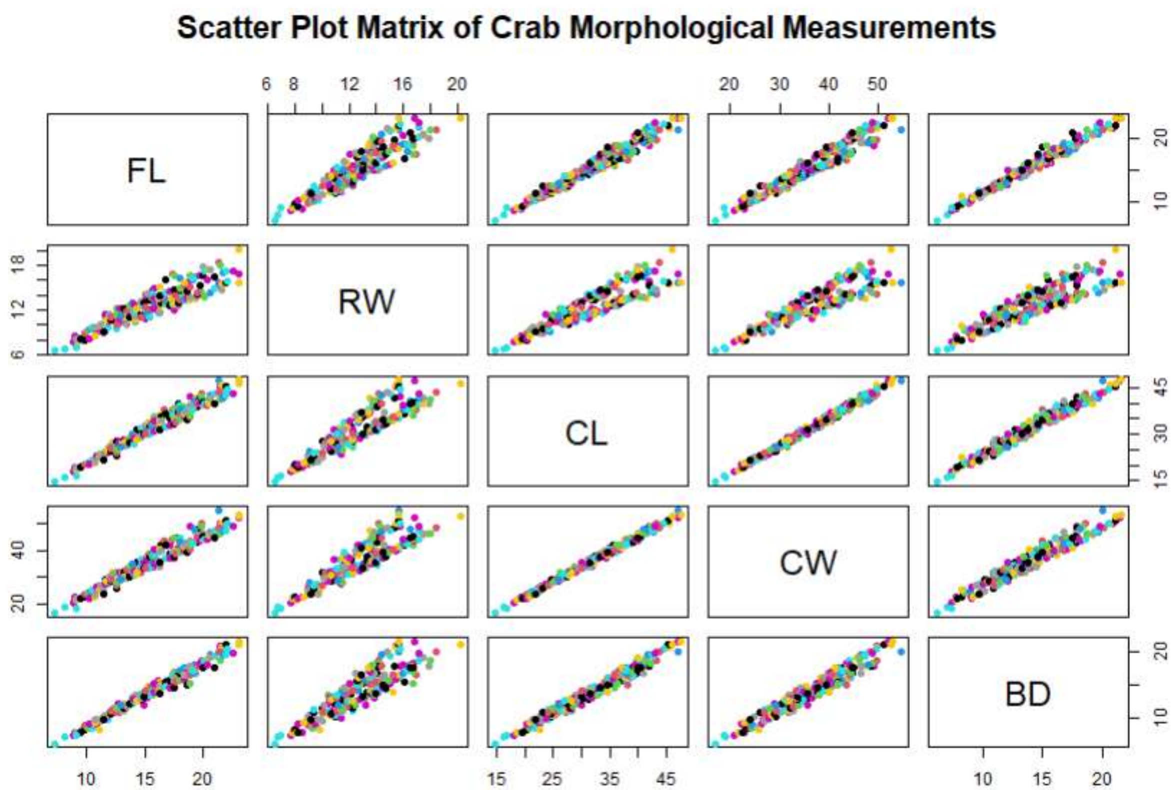
Additionally, the mean values (figure 1.3) for all five morphological measurements were calculated by species, sex, and the combination of species and sex. Across all measurements, species O crabs are larger than species B, and males are generally larger than females within each species. A grouped bar chart of the mean values shows that carapace width (CW) is the most distinctive feature for separating species, as it differs the most between B and O. At the same time, CW shows the least variation between sexes, meaning that male and female crabs within the same species are very similar in width. Other measurements, such as carapace length and frontal lobe size, also show differences, but these are less pronounced.

Figure 1.3



A scatter plot matrix of the five measurements (figure 1.4) reveals that all the traits are positively correlated. In other words, crabs that are larger in one dimension tend to be larger in the others. The relationships are strongest between carapace length and width, and between frontal lobe size and rear width, where the points follow a clear upward trend. Coloring the points by species highlights that species O crabs occupy the higher range of measurements, while species B crabs tend toward the lower end. This pattern indicates that crab body parts grow proportionally, and the overall size of a crab is reflected consistently across all measured dimensions.

Figure 1.4



In conclusion, the analysis of the crabs dataset highlights clear patterns in crab morphology. Species O crabs are larger than species B, and males are larger than females. Carapace width stands out as a key measurement for distinguishing species, while remaining relatively consistent between sexes. All five morphological measurements are closely related, showing proportional growth across the body. The dataset's balanced design allows for unbiased comparisons, and the combination of numerical summaries and visualizations provides a clear and coherent understanding of how species and sex influence crab size and shape.